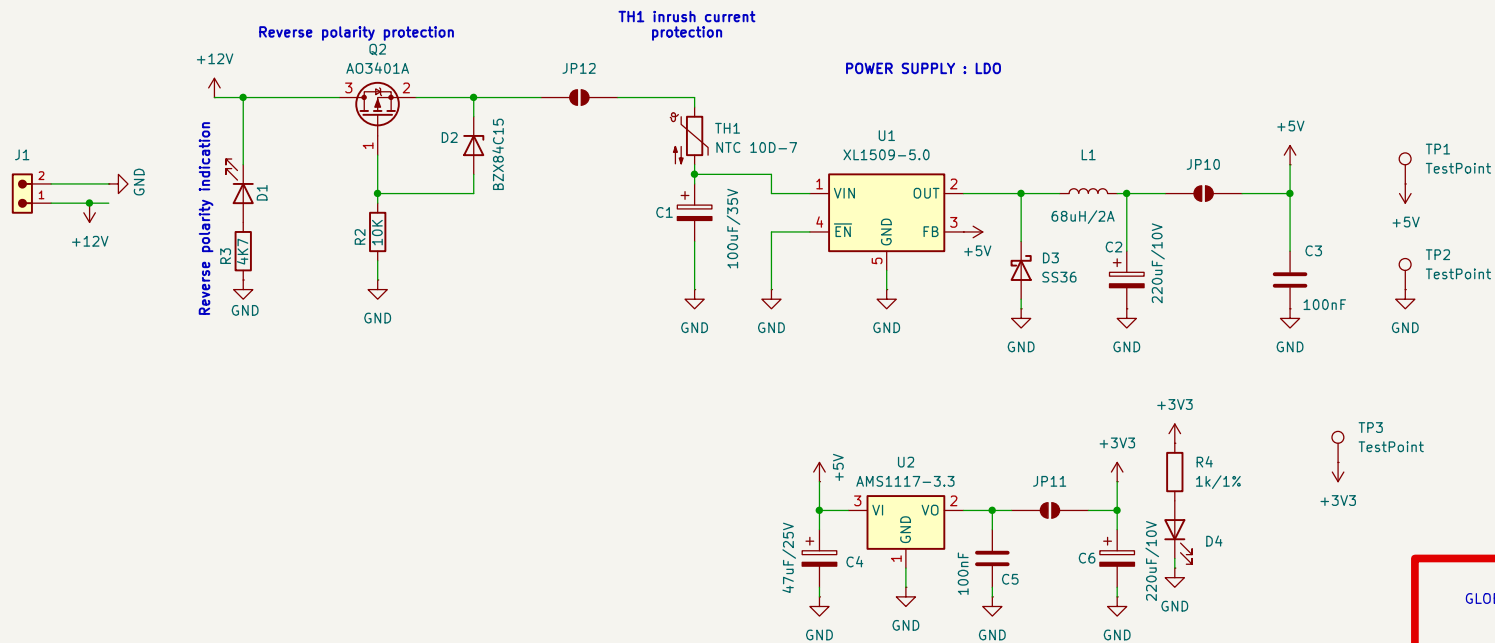






GLOBAL PINS / LABELS

→ +12V

→ +5V

→ +3V3

→ GND

Designed by: Lucas Tivana  
DC POWER SUPPLY  
Input DC 12/24V ->5V and 3V3 DC Output

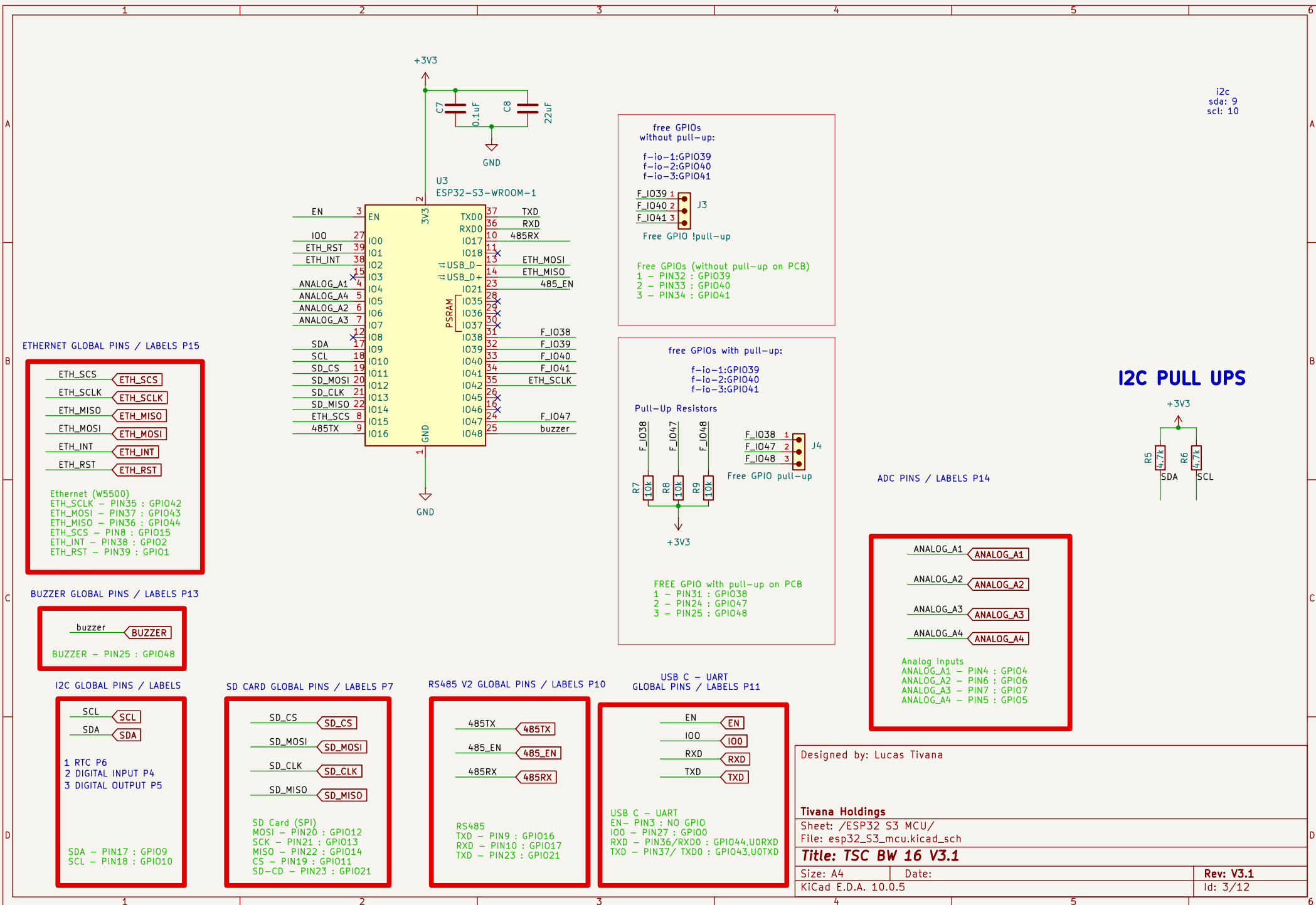
Tivana Holdings  
Sheet: /Power Supply/  
File: Power\_Supply.kicad\_sch

Title: TSC BW 16 V3.1

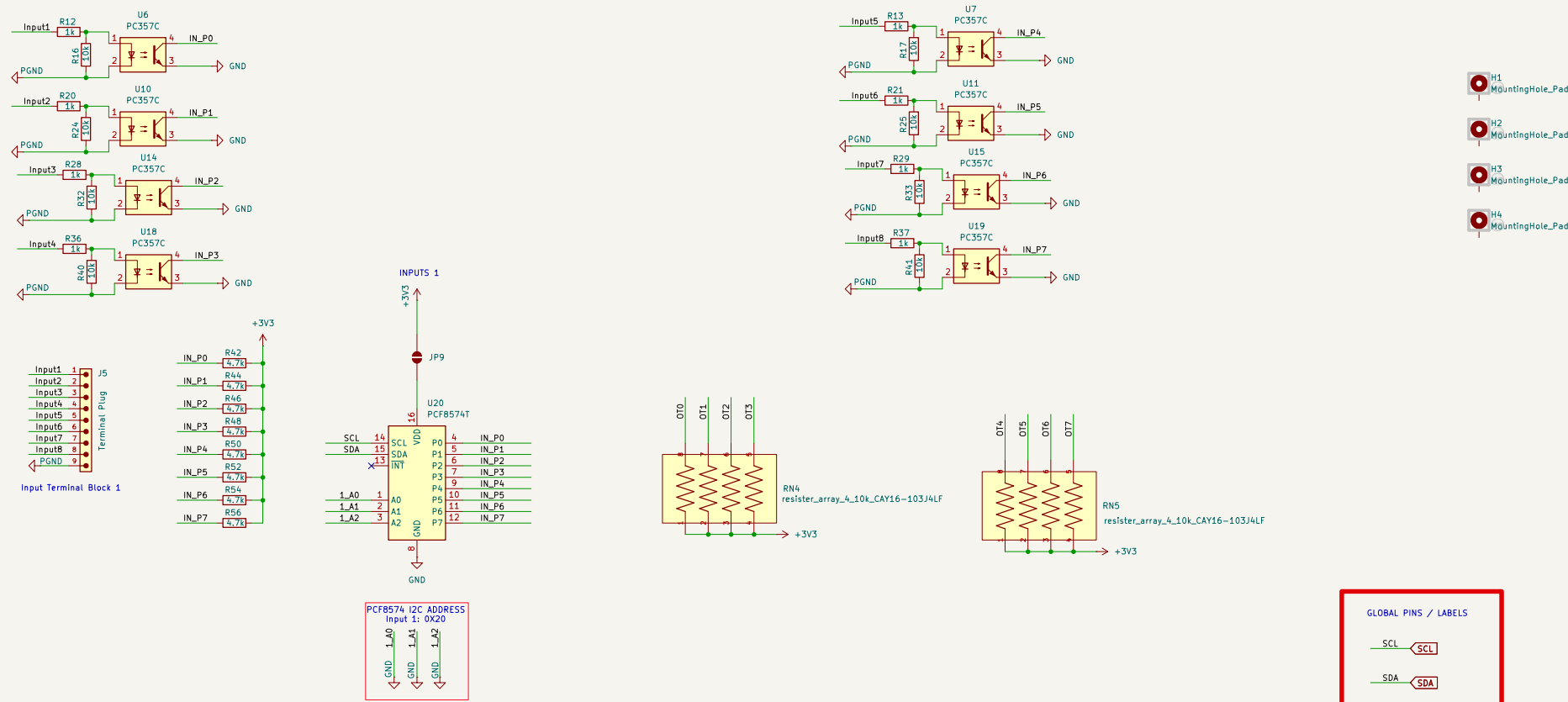
Size: A4  
KiCad E.D.A. 10.0.5

Date: 2026-04-20

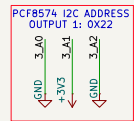
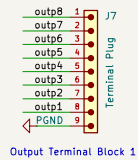
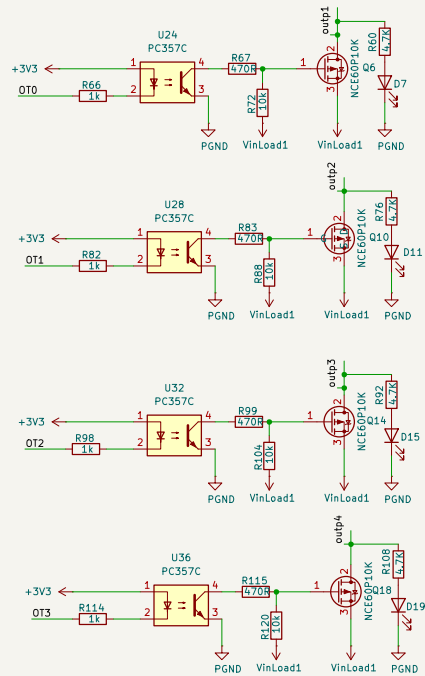
Rev: V3.1  
Id: 2/12



# 8 DIGITAL ISOLATED INPUTS

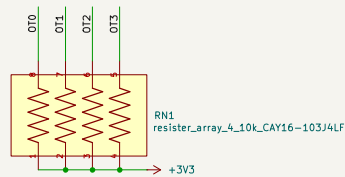
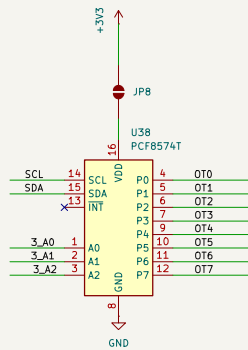


# 8 ISOLATED OUTPUTS

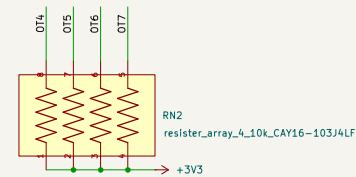
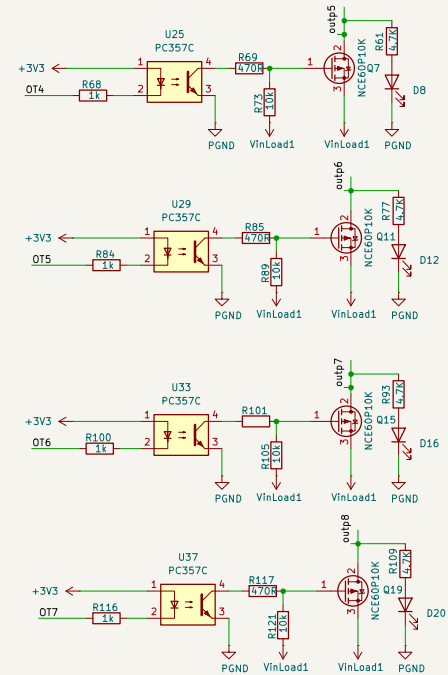


Output Terminal Block 1

GLOBAL PINS / LABELS



PCF Pull Up Resistors



Designed by: Lucas Tivana

Tivana Holdings  
Sheet: /8 DIGITAL OUTPUTS/  
File: 8\_digitalOutputs.kicad\_sch

Title: TSC BW 16 V3.1

Size: A3 Date: KICad E.D.A. 10.0.5

Rev: V3.1  
Id: 5/12

1 2 3 4 5

A

B

C

D

# RTC DS3231 & 24C04 EEPROM

24C04 I2C ADDRESS  
0x50 (A1 & A2 -> GND)

DS3231 I2C ADDRESS  
0x68 Fixed

GLOBAL PINS / LABELS

|     |     |
|-----|-----|
| SCL | SCL |
| SDA | SDA |

Designed by: Lucas Tivana

**Tivana Holdings**  
Sheet: /RTC/  
File: rtc.kicad\_sch

**Title: TSC BW 16 V3.1**

|                     |       |           |
|---------------------|-------|-----------|
| Size: A4            | Date: | Rev: V3.1 |
| KiCad E.D.A. 10.0.5 |       | Id: 6/12  |

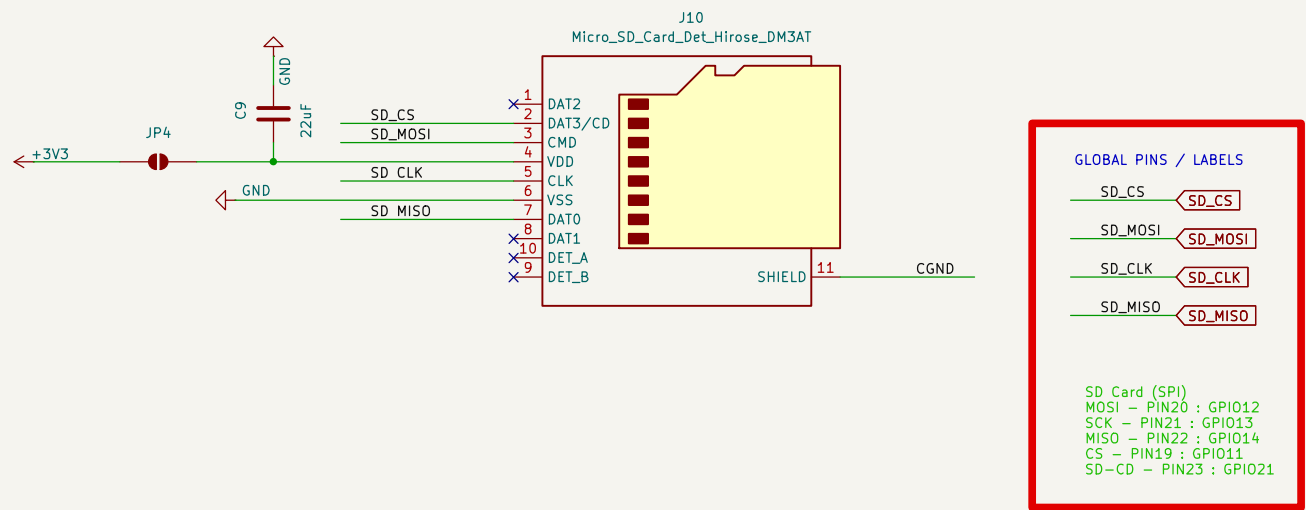
1 2 3 4 5

D

|                     |  |
|---------------------|--|
| SIZE: A1            |  |
| KiCad E.D.A. 10.0.5 |  |

Id: 6/12

# SD CARD



Designed by: Lucas Tivana

**Tivana Holdings**

Sheet: /SD CARD/

File: sd\_card.kicad\_sch

**Title: TSC BW 16 V3.1**

Size: A4

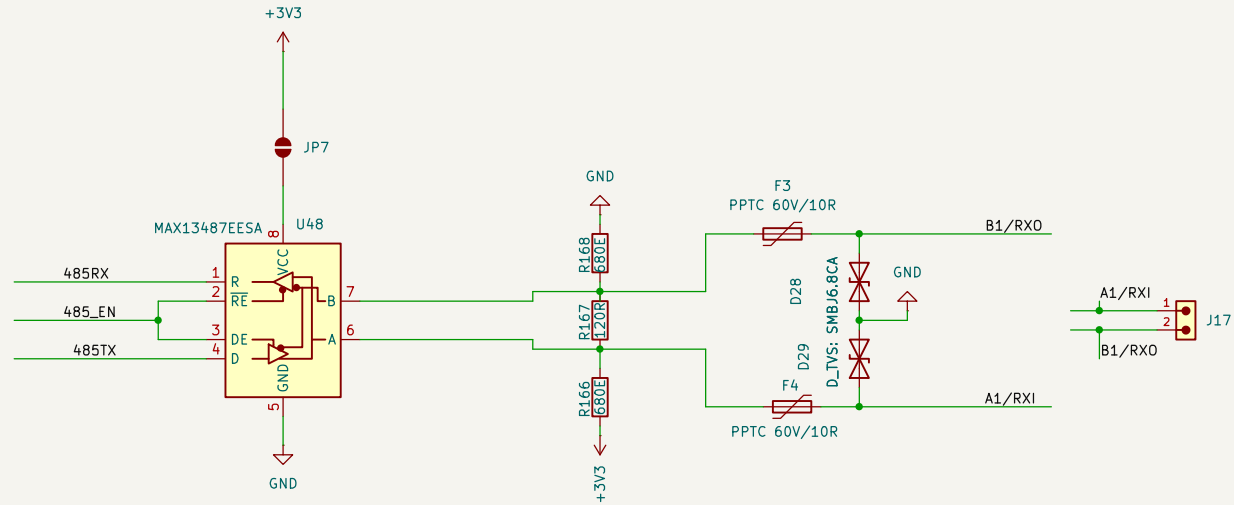
Date:

KiCad E.D.A. 10.0.5

**Rev: V3.1**

Id: 7/12

# RS 485 V2



```
#define RS485_EN 21

void rs485Transmit()
{
    digitalWrite(RS485_EN, HIGH);

    void rs485Receive()
    {
        digitalWrite(RS485_EN, LOW);
    }

    void sendRS485(String msg)
    {
        rs485Transmit();
        Serial2.print(msg);
        Serial2.flush();
        rs485Receive();
    }
}
```

## GLOBAL PINS / LABELS

485TX → 485TX  
485\_EN → 485\_EN  
485RX → 485RX

RS485  
TXD - PIN9 : GPIO16  
RXD - PIN10 : GPIO17  
TXD - PIN23 : GPIO21

Designed by: Lucas Tivana

## Tivana Holdings

Sheet: /RS485 v2/  
File: RS485 v2.kicad\_sch

**Title: TSC BW 16 V3.1**

Size: A4  
KiCad E.D.A. 10.0.5

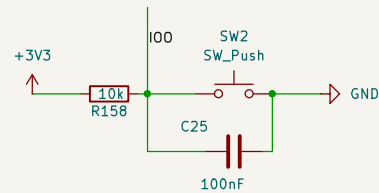
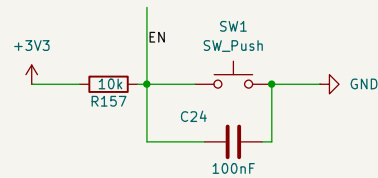
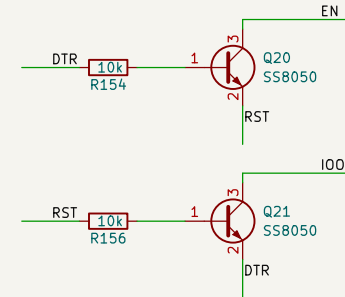
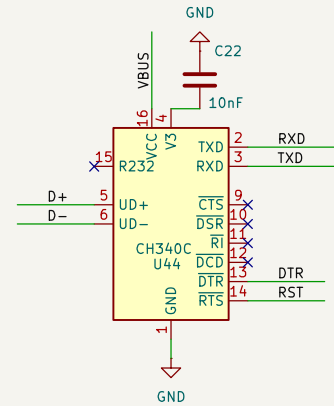
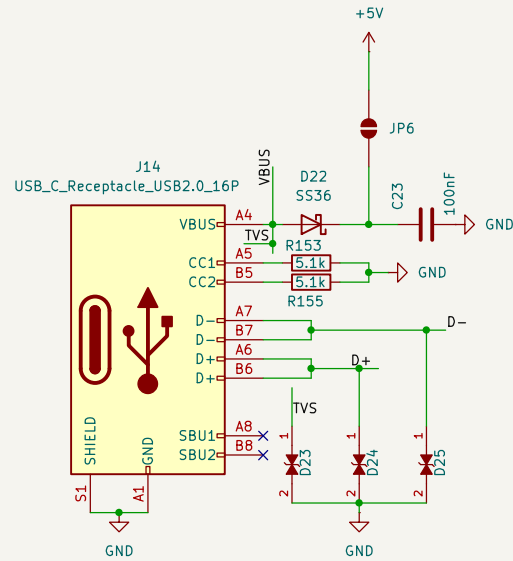
Date:

**Rev: V3.1**

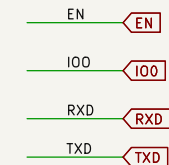
Id: 10/12

# USB – TO – UART

## AUTO PROGRAM



### GLOBAL PINS / LABELS



USB C – UART  
EN – PIN3 : NO GPIO  
IOO – PIN27 : GPIO0  
RXD – PIN36/RXD0 : GPIO44,U0RXD  
TXD – PIN37/ TXD0 : GPIO43,U0TXD

Designed by: Lucas Tivana  
USB Type C to UART  
Auto Program

**Tivana Holdings**

Sheet: /USB\_C to UART/  
File: usb\_c\_to\_uart.kicad\_sch

**Title: TSC BW 16 V3.1**

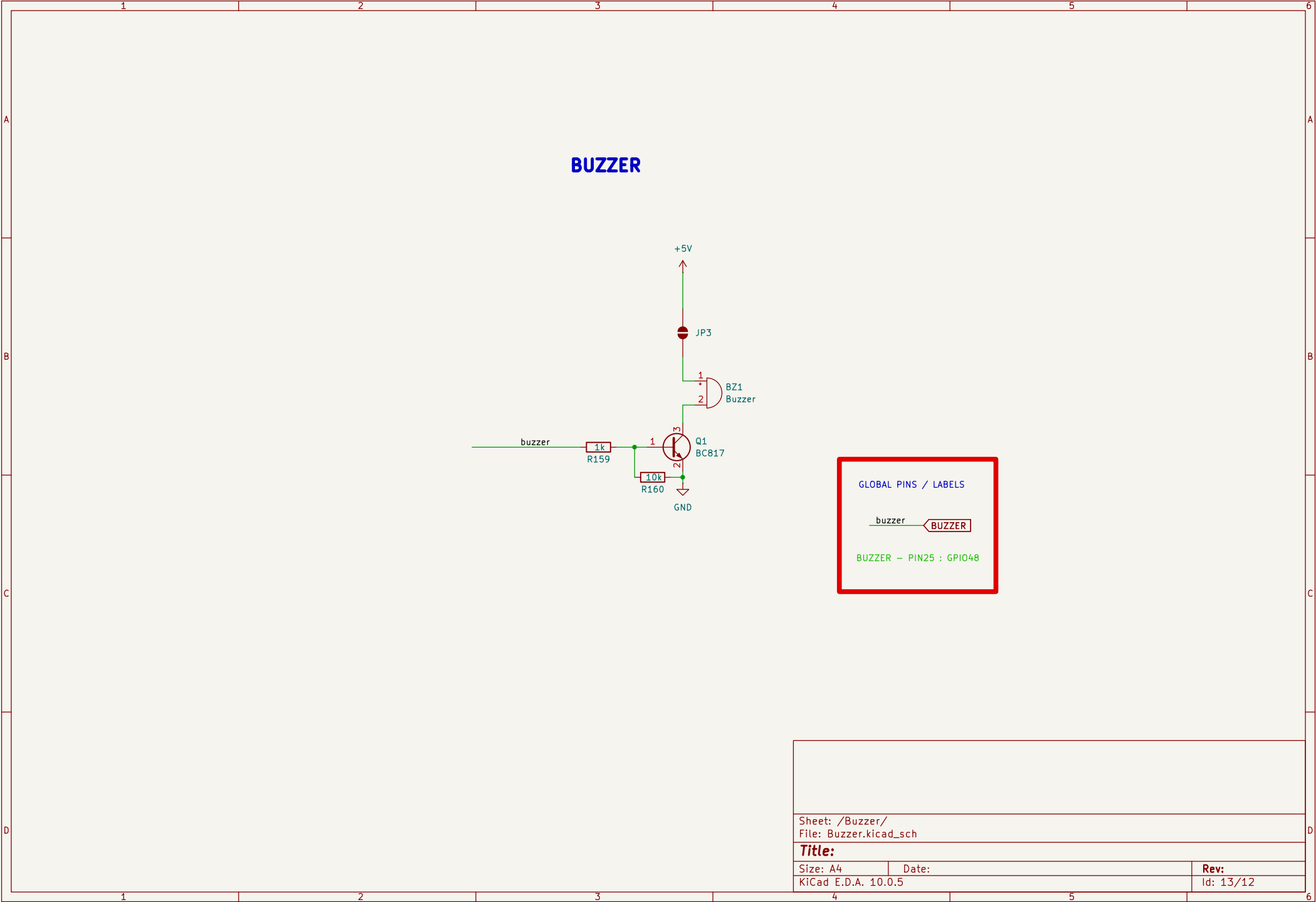
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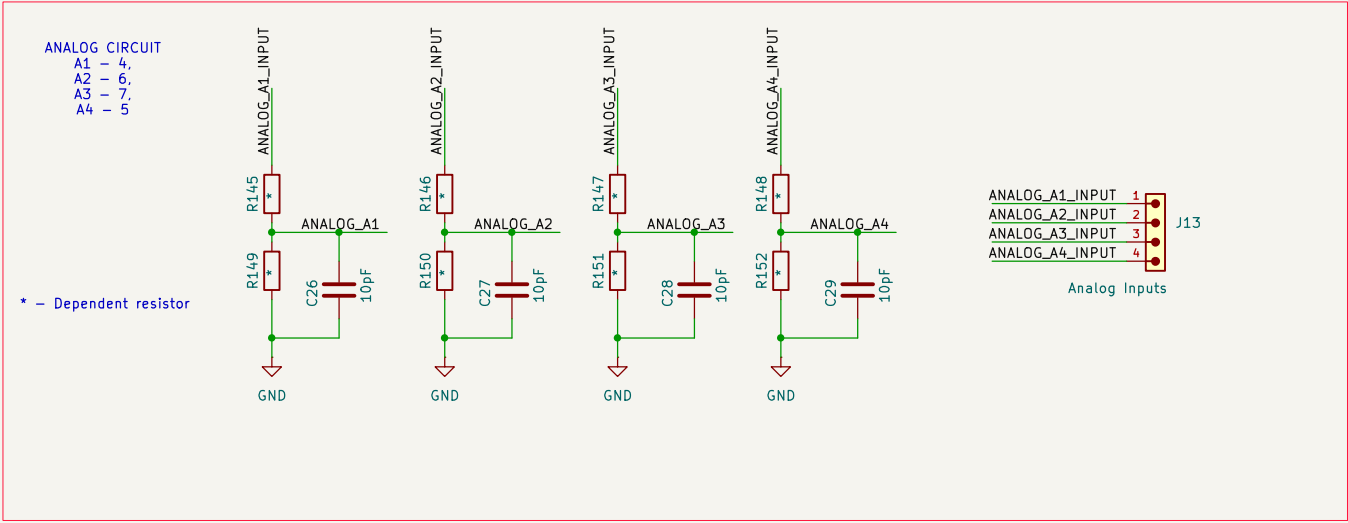
Date:

KiCad E.D.A. 10.0.5

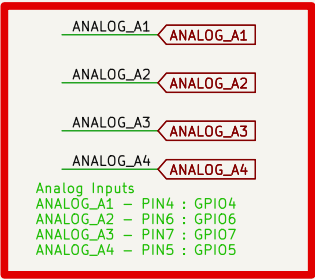
**Rev: V3.1**

Id: 11/12





ADC PINS / LABELS P14



Designed by: Lucas Tivana

**Tivana Holdings**

Sheet: /ADC PINS/  
File: adc\_pins.kicad\_sch

**Title: TSC BW 16 V3.1**

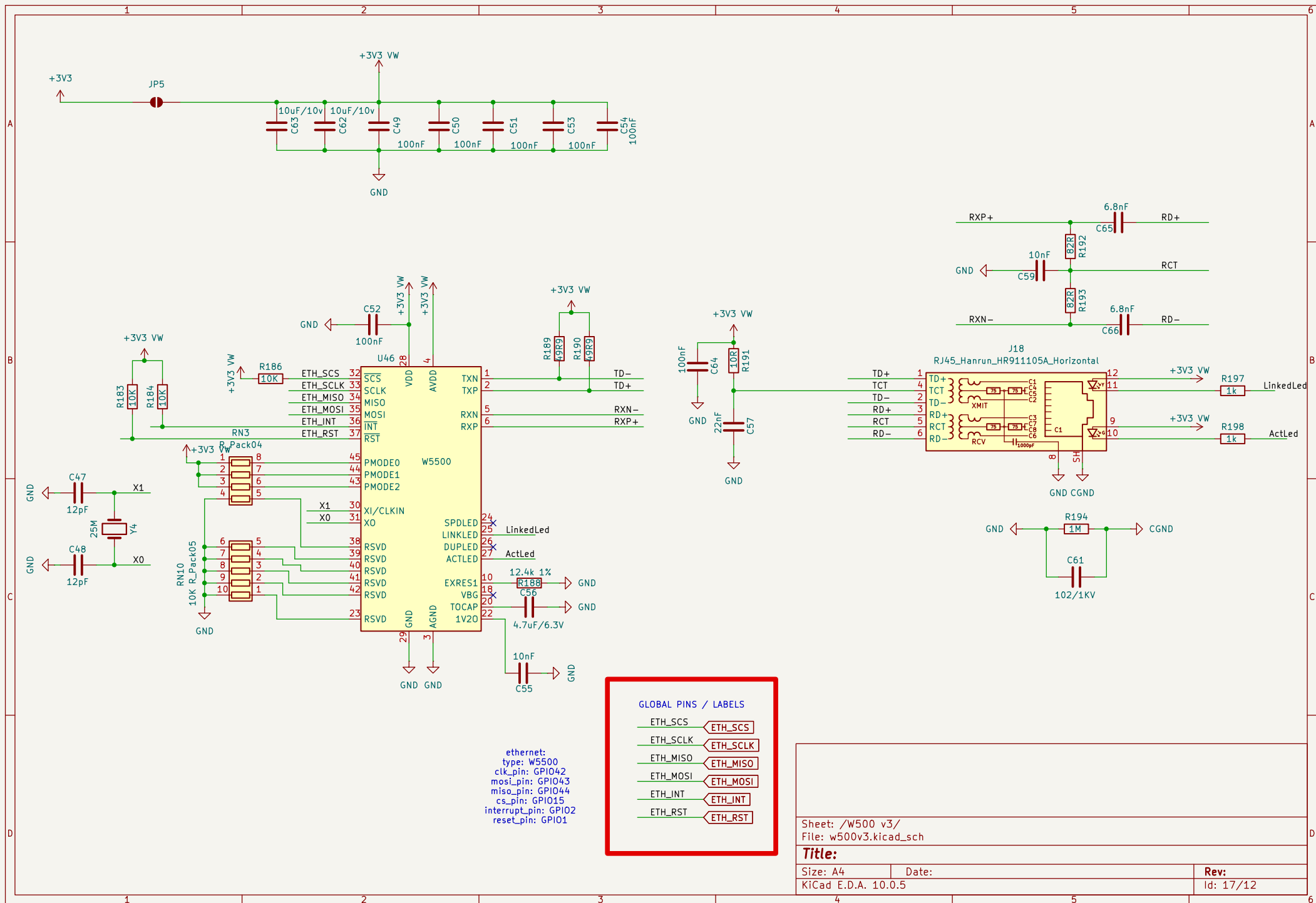
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Date:

KiCad E.D.A. 10.0.5

**Rev: V3.1**

Id: 14/12



Sheet: /w500 v3/  
 File: w500v3.kicad\_sch

**Title:**

Size: A4  
 KiCad E.D.A. 10.0.5

Date:

**Rev:**

Id: 17/12